Halward Inspection Services Ltd.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
HIS-2010-0318-DS-4-RA-104	Issued	2010-09-13

UT Thickness Data Sheet - 4-RA-104

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for the 4-RA-104 line located within the ATU-100. The scope of work involves establishing a baseline for the new 4-RA-104 spool as part of the broader 3-year regenerator overhead survey.

Kevin Doucette, acting as lead UT technician and SNT-TC-1A UT Level II for Halward Inspection Services Ltd., is responsible for the execution of the thickness measurements and the recording of the data.

Technical Specifications

The subject line, 4-RA-104, is an A106 Gr.B L/R exchanger rich side pipe with an outer diameter of 114.3 mm and a nominal wall thickness of 8.56 mm. The design limits for this circuit are 862 kPa(g) and 127 °C.

The minimum allowable thickness (t-min) for this line is 3.81 mm. Any reading recorded below this value shall be processed as an NCR.

Inspection Methodology

Thickness measurements were performed in accordance with ASTM E797 and the HIS-UT-002 Rev 1 procedure submitted for QSCP acceptance.

The inspection was further governed by QSCP-NDT-001 Rev 1. This requires the use of gauges with waveform display, the establishment of CMLs at fittings, and the use of scanning rather than spot measurements at elbows and tees.

Data Integration

The results contained herein are integrated into the HIS-2010-0318 report, which covers the baseline of the new 4-RA-104 spool and the general regenerator overhead survey.

Remaining life and future inspection intervals for the 4-RA-104 line will be calculated based on the larger of the long-term or short-term corrosion rates, with the next reading due at no more than half the remaining life.

Thickness readings, measured 2010-09-13

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2007-09-18	6.95	6.78	0.056	0.044	>50	2013-09-13	
3-RV-101-02	7.62	3.56	2007-09-18	6.89	6.59	0.068	0.075	40.6	2013-09-13	
3-RV-101-03	7.62	3.56	2007-09-18	6.56	6.16	0.094	0.090	27.8	2013-09-13	
3-RV-101-04	7.62	3.56	2007-09-18	6.66	6.43	0.080	0.048	35.9	2013-09-13	
3-RV-101-05	7.62	3.56	2007-09-18	6.87	6.63	0.072	0.084	36.8	2013-09-13	
3-RV-101-06	7.62	3.56	2007-09-18	6.71	6.43	0.080	0.068	35.9	2013-09-13	
3-RV-101-07	7.62	3.56	—	—	6.20	—	—	—	2013-09-13	
3-RV-101-08	7.62	3.56	—	—	6.71	—	—	—	2013-09-13	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2007-09-18	4.83	4.55	0.068	0.066	26.1	2013-09-13	
2-RV-102-02	5.54	2.79	2007-09-18	4.67	4.60	0.068	0.053	26.5	2013-09-13	
2-RV-102-03	5.54	2.79	2007-09-18	5.01	4.84	0.044	0.066	31.1	2013-09-13	
2-RV-102-04	5.54	2.79	2007-09-18	5.12	4.84	0.051	0.051	40.2	2013-09-13	
2-RV-102-05	5.54	2.79	—	—	5.04	—	—	—	2013-09-13	
2-RV-102-06	5.54	2.79	—	—	4.82	—	—	—	2013-09-13	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-03	8.56	3.81	—	—	8.59	—	—	—	2020-09-13	
4-RA-104-04	8.56	3.81	—	—	8.52	—	—	—	2020-09-13	
4-RA-104-05	8.56	3.81	—	—	8.43	—	—	—	2020-09-13	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2010-09-13	_____